Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-2430
Project Name	Team Enrollment Confirmation - FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how FitBuddy behaves under expected and peak load conditions. The testing plan below is tailored to the project's FastAPI backend, Gemini-powered generation, Jinja2 frontend, and SQLite data storage. Actual measured values are left as execution fields rather than being invented.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter / equivalent HTTP load-testing tool
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	/generate-workout, /submit-feedback, /view-all-users
Test Environment	Local FastAPI + Uvicorn server; SQLite database; Gemini API integration
Test Date	01 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal load: Generate personalized 7-day workout plan using /generate-workout.	10	60	Requests complete successfully with acceptable response time and no unexpected errors.
2	Concurrent load: Multiple users generate workout plans and nutrition tips.	25	120	Application remains stable; error rate stays within the target threshold.
3	Feedback load: Users submit feedback to update an existing plan through /submit-feedback.	15	90	Plans are updated correctly and database records remain consistent.
4	Spike load: Sudden increase in requests to the workout-generation endpoint.	50	60	Server recovers without crashes; requests are handled or fail gracefully.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	To be measured	Pending	Record average after load-test execution.
2	Response Time (Max)	< 5 seconds	To be measured	Pending	Record maximum response time.
3	Throughput (Req/sec)	To be measured	To be measured	Pending	Capture requests handled per second.
4	Error Rate	< 1%	To be measured	Pending	Check failed HTTP requests/API failures.
5	CPU Utilization	< 80%	To be measured	Pending	Monitor during peak test.
6	Memory Utilization	< 80%	To be measured	Pending	Monitor during peak test.

Step 4: Observations & Analysis

Key Findings	The planned tests focus on the major FitBuddy user flows: workout generation, feedback-based plan updates, and admin/user-plan retrieval. The application should be evaluated for response time, throughput, error rate, CPU usage, and memory usage under increasing concurrent requests.
Bottlenecks Identified	Potential areas to observe during execution include Gemini API response latency, concurrent FastAPI request handling, SQLite access under concurrent activity, and resource usage during spike load.
Optimization Steps Taken	Use asynchronous FastAPI request handling where applicable, keep database operations efficient, reuse a clean modular route structure, and monitor Gemini API calls and server resources during load testing. Record any actual optimization after test execution.

Step 5: Screenshots / Evidence

Screenshot 1	Insert JMeter/load-test summary or response-time graph here.
Screenshot 2	Insert CPU/memory/throughput or API test evidence here.

Team Work Allocation

The responsibilities are assigned in the same order as the team-member names provided, and aligned with the project workflow documented for FitBuddy.

S.No	Member	Assigned Area	Work / Responsibility
1	Divya.D	Milestone 1 – Model Selection & Architecture	Research/select Gemini model, define application architecture, and support development-environment setup.
2	Deepika.M	Milestone 2 – Core Functionalities & FastAPI Backend	Develop workout generation, nutrition-tip generation, feedback update, storage functions, and FastAPI input processing.
3	Dharani.T	Milestone 3 – routes.py Development	Implement and test the main routes: home, workout generation, feedback update, and admin view; connect backend modules.
4	Divya shree.R	Milestone 4 – Frontend Development	Design index.html/result.html/all_users.html, responsive CSS, and Jinja2 dynamic rendering.
5	DivyaDharshini.V	Milestone 5 – Deployment & Testing	Prepare local deployment, verify Uvicorn/FastAPI operation, execute performance testing, collect evidence, and document results.

Project Basis

FitBuddy is documented as a web-based application that uses FastAPI, Google Gemini models, Jinja2 templates, SQLAlchemy, and SQLite to generate personalized 7-day workout plans, nutrition/recovery tips, and feedback-based plan updates. The project also includes an admin view of users and plans.

Testing Note

The performance targets and scenarios above are prepared from the supplied project description and the provided Performance Testing template. Actual performance measurements (response time, throughput, error rate, CPU, and memory) must be filled in after the team runs the tests; this document does not claim unperformed measurements as actual results.